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UW24MultiToneSrcDefTopic

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About the UltraWave24 MWMultiToneSource

About the UltraWave24 MWMultiToneSource

The MWMultiToneSource is a programmable RF source that generates two-tone or multitone signals in the frequency range from 50MHz to 6GHz. Applications for the MWMultiToneSource

include IMD, IP2, and IP3 tests.

The following topics are available:

• [Features](#)

• [Safety Information](#)

• [Key Terms](#)

MWMultiToneSource information includes the following sections:

• [UltraWave24 MWMultiToneSource Theory of Operation](#)

• [UltraWave24 MWMultiToneSource Programming](#)

• [UltraWave24 MWMultiToneSource Debug Display](#)

Related Topics

• [UltraWave24 Specifications](#)

• [MWMultiToneSource \(VBT syntax\)](#)

• [UltraWave24 DIB Design Guidelines](#)

• [About the UltraWave24 Subsystem](#)

• [UltraWave24 Examples](#)

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About the UltraWave24 MWMultiToneSource : Features

Features

The MWMultiToneSource generates either two-tone signals by adding two CW signals or more complex signals by adding a modulated signal to a CW signal or by adding two modulated signals. Basic functional specifications are:

- Added signals components can be CW or modulated
- CW signal frequency range: 50MHz to 6GHz
- Modulated signal frequency range: 76MHz to 6GHz
- CW amplitude range (frequency dependent):
 - Tone 1 signal path: -130dBm to +5dBm
 - Tone 2 signal path: -130dBm to +5dBm

For UltraWave24 hardware specifications, see [UltraWave24 Specifications](#).

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About the [UltraWave24 MWMultiToneSource](#) : Safety Information

Safety Information

When operating the Microwave instruments, always follow general [UltraFLEX](#) test system safety guidelines. See [Test System Safety](#).

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About the UltraWave24 MWMultiToneSource : Key Terms

Key Terms

add path	Signal combiner allowing two RF signals to be added to create a two-tone signal or a more complex multitone signal if one or both component signals is modulated.
multitone	A signal composed of two or more discrete frequencies. For the MWMultiToneSource, a multitone is created by adding two signals, each of which can be a CW or modulated signal.

For more MW terms and definitions, see [Key Terms](#) in About the UltraWave24 Subsystem.

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UltraWave24 MWMultiToneSource Debug Display

UltraWave24 MWMultiToneSource Debug Display

The UltraWave24 Microwave Debug Display can display one or more instrument blocks for the MWMultiToneSource logical instrument, each showing the state of a MWMultiToneSource in the context of a site.

This section includes the following topic:

- [UltraWave24 MWMultiToneSource Debug Display Controls](#)

For an overview of all displays supporting the UltraWave24 subsystem, see [UltraWave24 Displays](#). For more information on using debug displays, see the following:

- [Using Debug Displays in TDE](#)

- [Producing Code from Debug Displays](#)

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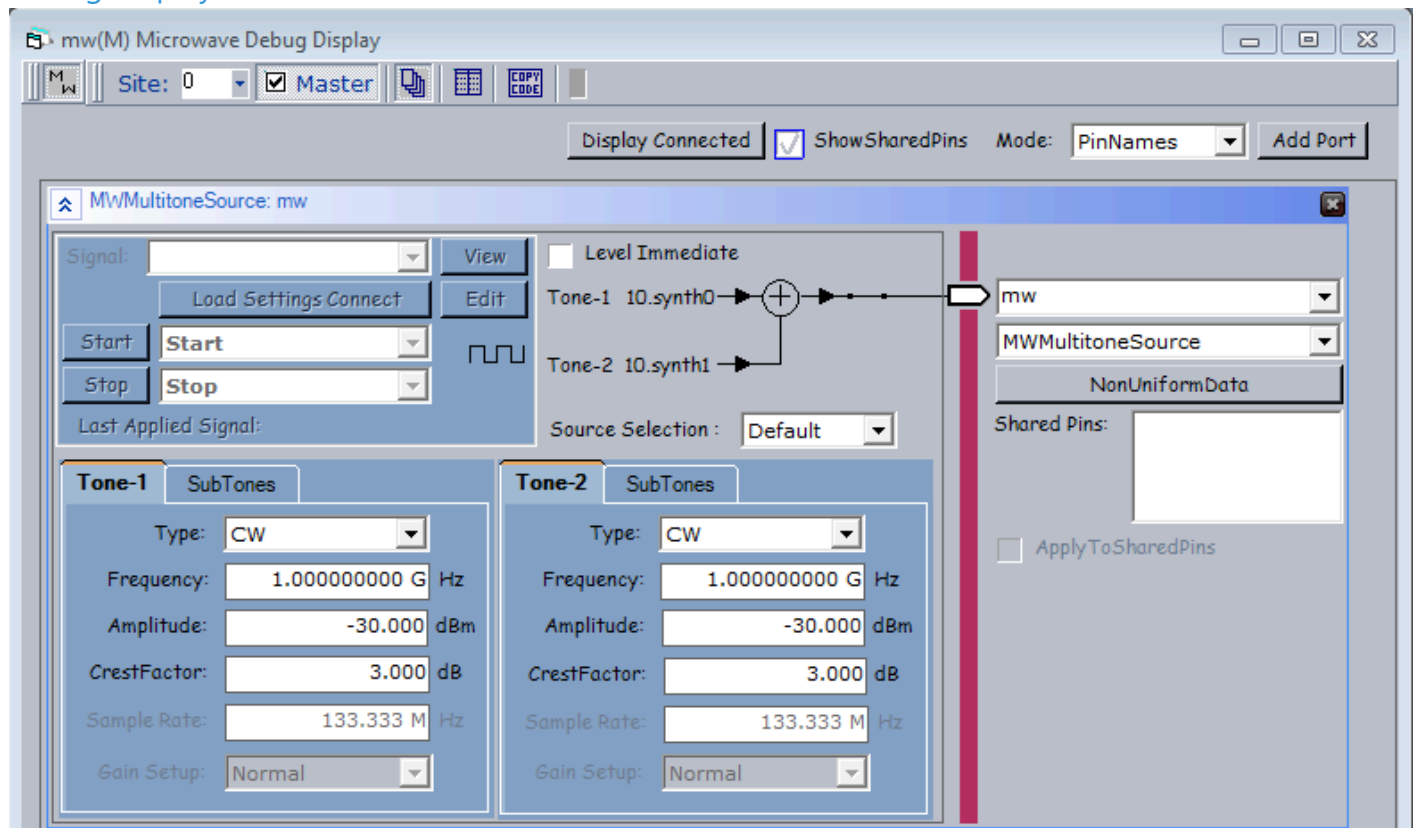


[UltraWave24 MWMultiToneSource Debug Display](#) : UltraWave24 MWMultiToneSource Debug Display Controls

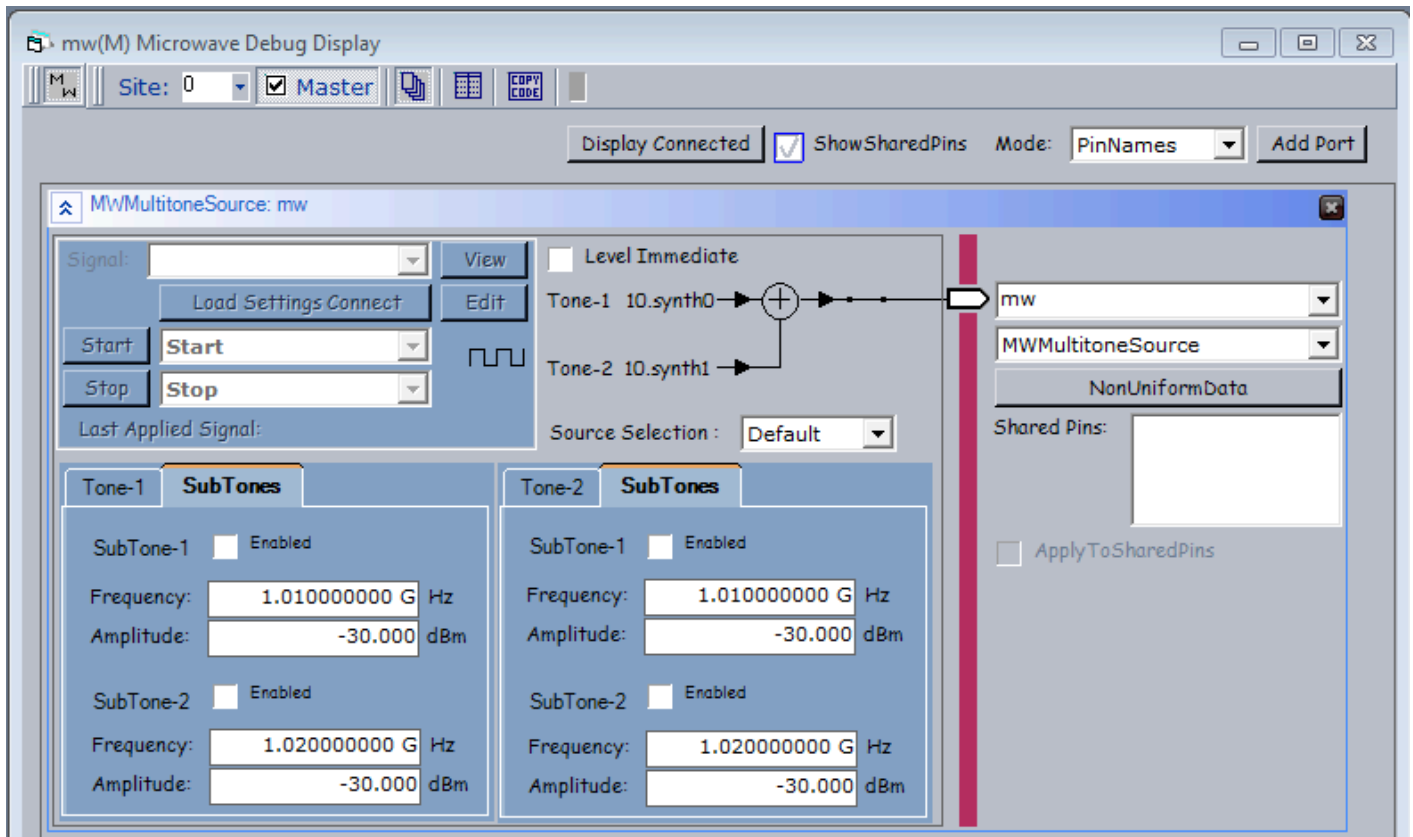
UltraWave24 MWMultiToneSource Debug Display Controls

The display controls for the MWMultiToneSource assigned to a port allow you to view and set instrument parameters, which are also accessible through the Pins interface of the MWMultiToneSource VBT language.

For information about the common debug display controls, see [UltraWave24 Microwave Board Debug Display](#).



MWMultiToneSource Logical Instrument Display, Tone Tabs Selected



MWMultiToneSource Logical Instrument Display, SubTones Tabs Selected

Tone1 always controls the primary signal path. For the example display above, Tone1 is a modulated signal and Tone2 is a CW signal.

The following controls are available on the MWMultiToneSource logical instrument block:

Signal	Name of the signal used by the source. Available signals appear on the drop-down list. If a default signal has been defined, the default signal's name appears automatically in this field. If the menu contains no signal names, no signals were previously defined in VBT. The MWMultiToneSource can operate without a signal.
View	Displays a list of signal viewer tools (such as WaveScope or VectorScope) for the selected signal on a drop-down list of available viewers.
Load Settings Connect	Loads the selected signal's timing settings as specified on the Mixed Signal Timing sheet, as well as all of the instrument's settings and connects the MWMultiToneSource.
Edit	Brings up the PSet and Signals Editor (PSSE) with the selected signal loaded. This control is grayed out if no signals are available. See PSet and Signals Editor in the TDE section.

Info	Displays information for the selected signal in a pop-up balloon window.
Start	Starts sourcing the selected signal using one of the start options: - Start - Start1 - StartE - Start1E
Stop	Stops sourcing the selected signal using one of the stop options: - Stop - StopE
	The square wave is a sourcing indicator that moves when the source is sourcing on the tester. This feature only works on line.
Last Applied Signal	Displays the name of the signal last loaded to the MWMultiToneSource.
SourceSelection	Displays and sets the synthesizer used to generate modulated source signals, Primary, Secondary, or Default.
synth label	Identifies the two synthesizers (slot and number) used by this MWMultiToneSource logical instrument for Tone-1 (primary) and Tone-2 (secondary).
connection relay	The relay control connects or disconnects the MWMultiToneSource logical instrument to the specified pin or channel and displays the connection state. Click the relay to open or close the connection.
Level Immediate	Sets the MWMultiToneSource leveling mode to Level Immediate, when checked. See MWMultiToneSource.Pins.LevelImmediate in the VBT Syntax Reference.
Tone-1 and Tone-2 tabs	
Type	Displays and sets the type of modulation on a drop-down list:

- **CW**: Grays out Tone parameters (Sample Rate and Gain Setup).
- **Modulated**
- **Disabled**: Grays out Tone parameters (Sample Rate and Gain Setup) and SubTones parameters on the associated tab.

Frequency	Displays and sets the signal frequency. The display has separate frequencies for Tone-1 and Tone-2. The tooltip displays the allowable frequency range.
Amplitude	Displays and sets the signal amplitude. The display has separate amplitudes for Tone-1 and Tone-2. The tooltip displays the allowable amplitude range.
CrestFactor	Displays and set the crest factor for a tone. The display has separate crest factors for Tone-1 and Tone-2. The tooltip displays the allowable range.
Sample Rate	Displays and sets the modulator sample rate for a modulated tone. Enabled only if Modulation is selected. The tooltip displays the allowable sample rate range.
Gain Setup	Displays and sets the gain setup for a modulated tone. For the UltraWave24 MWMultiToneSource, only Normal gain is used.
Sub-Tones tab	Allows you to enable and set subtone parameters, Frequency and Amplitude, for SubTone-1 and SubTone-2 for each tone.

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UltraWave24 MWMultiToneSource Programming

UltraWave24 MWMultiToneSource Programming

The MWMultiToneSource is a MW logical instrument in the IG-XL environment. In general, you program the MWMultiToneSource like the other MW instruments.

The following topics are available:

- Sourcing Two-Tone Signals Using the MWMultiToneSource
- MWMultiToneSource Pattern Control

Before writing a program using the MWMultiToneSource, see [UltraWave24 Instrument Programming](#).

For more information, see [MWMultiToneSource](#) in the VBT Syntax Reference.

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UltraWave24 [MWMultiToneSource Programming](#) : Sourcing Two-Tone Signals Using the MWMultiToneSource

[Sourcing Two-Tone Signals Using the MWMultiToneSource](#)
The following topics are available:

- | | |
|---|--------------------------------------|
| • | Programming Concepts |
| • | Programming Steps |
| • | Examples |

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UltraWave24 MWMultiToneSource Programming : Sourcing Two-Tone Signals Using the MWMultiToneSource : Programming Concepts

Programming Concepts

Source Level Calibration

Source level calibration, also called leveling, finely calibrates the MWMultiToneSource at each frequency set in a program. Leveling takes place at run time, and each tone (the tone and the subtone) is leveled independently.

Leveling occurs the first time a specific frequency is set for an individual tone. Later VBT statements setting the same frequency use cached data and do not level, unless the program invokes leveling using the Pins.LevelImmediate VBT property.

For more information, see TheHdw.MWMultiToneSource.Pins.LevelImmediate.

Using Subtones

The UltraWave24 allows you to generate subtones, digitally generated CW tones that are summed with the standard CW or modulated signals from the UltraWave 24 sources. Subtones are presented under the MWMultiToneSource logical instrument.

VBT includes a .Subtones interface in TheHdw.MWMultiToneSource.Pins.Tones that allows you to:

- | | |
|--|---|
| | <ul style="list-style-type: none">Program up to two subtones for each tone |
| | <ul style="list-style-type: none">Independently enable and control each subtone |

You can also disable either of the RF synthesizers using the .Tones.Enabled node. The default is not use subtones and to use two RF synthesizers.

For more information, see [Creating Subtones](#) in the Theory of Operation section and [TheHwd.MWMultiToneSource.Pins.Signals.Item.Tones.Item.Subtones](#) in the VBT Syntax Reference.

TSD File Support

The MWMultiToneSource accepts signal definitions for modulated signals in TSD files. TSD files are loaded using the .SignalDefinitionFile property.

For more information, see:

- Using the ESA Toolkit
- TheHdw.MWMultiToneSource.Pins.Signals.Item.Tones.Item.SignalDefinitionFile

MWMultiToneSource Reset Values

The table lists the local reset values for the MWMultiToneSource.

MWMultiToneSource Reset Values

Setting	Reset Value
Amplitude (Tones(1))	-30dBm
Frequency (Tones(1))	1GHz
Modulation	Disabled
Amplitude 2 (Tones(2))	-30dBm
Frequency 2 (Tones(2))	1GHz
Modulation	Disabled
Connections	Disconnected

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UltraWave24 MWMultiToneSource Programming : Sourcing Two-Tone Signals Using the MWMultiToneSource : Programming Steps

Programming Steps

The MWMultiToneSource can produce a two-tone signal (two CW signals), each with a programmed frequency and amplitude. The MWMultiToneSource can also produce a multitone by combining two modulated signals or by combining a CW signal with a modulated signal. To source a two-tone signal to an assigned RF port:

1. Set source signal amplitude and frequency for tone 1 using the Tones interface.
2. Set source signal amplitude and frequency for tone 2 using the Tones interface.
3. Connect to the DUT pin or channel.

To source a multitone signal created by combining two modulated signals to an assigned RF port:

1. Add a new Signal to the MWMultiToneSource collection.
2. Set Signal parameters for tone 1 using the Tones interface.
3. Set Signal parameters for tone 2 using the Tones interface.
4. Turn on modulation for both Signals.
5. Connect to the DUT pin or channel using LoadSettingsConnect.

6. [Start the Signal or Signals using a VBT statement or by running a pattern.](#)

To source a tritone signal to an assigned RF port:

1. [Set source signal amplitude and frequency for tone 1 using the .Tones interface.](#)

2. [Set source signal amplitude and frequency for subtone 1 of tone 1 using the .Tones.Subtones interface.](#)

3. [Set source signal amplitude and frequency for subtone 2 of tone 1 using the .Tones.Subtones interface.](#)

4. [Disable tone 2.](#)

5. [Connect to the DUT pin or channel.](#)

[See Examples.](#)

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UltraWave24 MWMultiToneSource Programming : Sourcing Two-Tone Signals Using the MWMultiToneSource : Examples

Examples

The following examples are available:

- Two-Tone Signal
- Multitone Signal from Two Modulated Signals
- Tritone Signal

Two-Tone Signal

This example code function accepts two separate frequencies (tones) and a single amplitude (RF power) for both.

```
Public Function MWMultiToneSourceSignal(ByVal Source_Pin As String, _
                                       ByVal F1_freq As Double, _
                                       ByVal F2_freq As Double, _
                                       ByVal RF_power As Double) As Long
```

```
Dim F1_power_level As Double
```

```
Dim F2_power_level As Double
```

```
' Set up low-side tone F1 using MW source1
```

```
F1_power_level = RF_power
```

```
With TheHdw.MWMultiToneSource.Pins(Source_Pin).Tones(1)
```

```
.Frequency = F1_freq
.Amplitude = F1_power_level
End With
```

```
' Set up high-side tone F2 using MW source2
F2_power_level = RF_power
```

```
With TheHdw.MWMultiToneSource.Pins(Source_Pin).Tones(2)
    .Frequency = F2_freq
    .Amplitude = F2_power_level
End With
```

```
' Connect the multitone source
TheHdw.MWMultiToneSource.Pins(Source_Pin).Connect
```

```
End Function
```

Multitone Signal from Two Modulated Signals

This example code function uses Gen4 waveforms to create a multitone signal. The function accepts frequency, amplitude (level), waveform file, IF frequency (required for Gen4 waveforms), and other Signal parameters for each modulated "tone."

```
Public Function SourceMultiTone(ByVal rf_freq As Double, _
    ByVal rf_freq1 As Double, _
    ByVal rf_level As Double, _
    ByVal rf_level1 As Double, _
    ByVal modsrc_segname As String, _
    ByVal modsrc_segname1 As String, _
    ByVal SigName As String
    ByVal IfFreqOne As Double, _
    ByVal IfFreqTwo As Double, _
    ByVal sampleRateOne As Double, _
    ByVal sampleRateTwo As Double, _
    ByVal sampleSzOne As Long, _
    ByVal sampleSzTwo As Long) As Long
```

```
' Add a new Signal to the MWMultiToneSource Signals collection and set
' all Signal parameters to default values.
```

```
TheHdw.MWMultiToneSource.Pins("RF_PIN").Signals.Add (SigName)
```

```
' Set the name of the Signal used by default when a Signal is applied
```

' without a Signal name.

```
TheHdw.MWMultiToneSource.Pins("RF_PIN").Signals.DefaultSignal = SigName
```

' If using Gen4 waveforms, IF Frequency must be specified.

' The UltraWave24 MWMultiToneSource uses IFFrequency as input parameters

' to convert the IF data to IQ data before loading it into the AWG.

```
TheHdw.MWMultiToneSource.Pins("RF_PIN").Signals(SigName) _  
    .Tones(1).IFFrequency = IfFreqOne
```

```
TheHdw.MWMultiToneSource.Pins("RF_PIN").Signals(SigName).Tones(2). _  
    IFFrequency = IfFreqTwo
```

' Set up parameters for first signal using Gen4 waveforms.

```
With TheHdw.MWMultiToneSource("RF_PIN").Signals(SigName).Tones(1)  
    .SampleRate = sampleRateOne  
    .SampleSize = sampleSzOne  
    .Type = tLMWSignalTypeIF  
    .ReverseSpectrum = True  
    .Frequency = rf_freq  
    .CrestFactor = 3.1  
    .Amplitude = rf_level  
    .WaveDefinitionName = modsrc_segname
```

End With

' Set up parameters for second signal using Gen4 waveforms

```
With TheHdw.MWMultiToneSource("RF_PIN").Signals(SigName).Tones(2)  
    .SampleRate = sampleRateTwo  
    .SampleSize = sampleSzTwo  
    .Type = tLMWMTSignalTypeIF  
    .ReverseSpectrum = True  
    .Frequency = rf_freqI  
    .CrestFactor = 3.2  
    .Amplitude = rf_levelI  
    .WaveDefinitionName = modsrc_segnameI
```

End With

' Turn on modulation for both signals

```
TheHdw.MWMultiToneSource("RF_PIN").Tones(1).Modulated = True
```

```
TheHdw.MWMultiToneSource("RF_PIN").Tones(2).Modulated = True
```

' Add site loop here if needed

' Set up and connect the MWMultiToneSource

TheHdw.MWMultiToneSource.Pins("RF_PIN").Signals(SigName).LoadSettingsConnect

TheHdw.Wait (0.005)

'Start the signal from VBT or run a pattern to start the signal

TheHdw.MWMultiToneSource("RF_PIN").Signals(SigName).Start (tlStartImmediately)

'
'
'
'

' Disconnect the MWMultiToneSource

TheHdw.MWMultiToneSource.Pins("RF_PIN").Disconnect

End Function

Tritone Signal

This example code function generates three tones of equal power:

Public Function SourceTriTone(ByVal Source_Pin As String, _
ByVal F1_freq As Double, _
ByVal Sub1_freq As Double, _
ByVal Sub2_freq As Double, _
ByVal RF_power As Double) As Long

With TheHdw.MWMultiToneSource.Pins(Source_Pin).Tones(1)

' Set up CW synthesizer

.Frequency = F1_freq

.Amplitude = RF_power

' Enable and set up subtone 1

' Frequency must be within the modulation bandwidth of the parent Tone

.Subtones(1).Enabled = True

.Subtones(1).Frequency = Sub1_freq

.Subtones(1).Amplitude = RF_power

' Enable and set up subtone 2

' Frequency must be within the modulation bandwidth of the parent Tone

.Subtones(2).Enabled = True

```
.Subtones(2).Frequency = Sub2_freq
.Subtones(2).Amplitude = RF_power
```

End With

' Disable tone 2

```
TheHdw.MWMultiToneSource.Pins(Source_Pin).Tones(2).Enabled = False
```

' Connect the multitone source

```
TheHdw.MWMultiToneSource.Pins(Source_Pin).Connect
```

End Function

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UltraWave24 MWMultiToneSource Programming : MWMultiToneSource Pattern Control

MWMultiToneSource Pattern Control

You can control an MWMultiToneSource instrument using pattern microcodes, allowing changes to instrument states and parameters during a program. You can obtain repeatable timing by programming an instrument using pattern microcodes and Signals programming.

You create a pattern independently of the test program using the PatternTool or a text editor.

Patterns are composed of a series of numbered digital vectors. These digital vectors define the data for each channel listed in the pattern, as well the pattern microcode used for controlling the instrument during pattern execution.

The following topics are available:

- MWMultiToneSource Pattern Microcodes
- MWMultiToneSource ASCII Pattern Syntax

Note:

To use pattern microcodes to control an MWMultiToneSource on UltraFLEX, you must include a digital pin in the pin map, the channel map, and the pin list of the pattern module.

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UltraWave24 MWMultiToneSource Programming : MWMultiToneSource Pattern Control : MWMultiToneSource Pattern Microcodes

MWMultiToneSource Pattern Microcodes

A microcode controls a single function, such a starting or stopping the sourcing of a signal. Extended microcode commands are available for use inside a pattern. Patterns allow you to control the precise timing of when a signal is sourced. To establish repeatable timing, or to synchronize with other instruments (such as the digital subsystem) in the tester, use a pattern. If you want to start generating a signal but are not concerned with precise timing, however, then you do not need a pattern.

The table lists the extended microcodes used to issue commands to an MWMultiToneSource instrument from a pattern.

MWMultiToneSource Pattern Microcodes

Microcode	Functional Description
Resync	Forces phase alignment of an MWMultiToneSource instrument clock with the HSD T0 clock. Resync is also required to align MWModulatedSources to each other (if time alignment is necessary). See Use of Resync.
Start <SignalName>	The instrument immediately (on the next available logical analog clock) starts sourcing the named signal in SMEM and repeats it continuously until a Stop microcode is executed, the instrument is reset, or another signal is started. This microcode takes an optional argument, which is the name of the signal to be sourced. If the signal name is not specified, the

	default signal is used (see TheHdw.MWMultiToneSource.Pins.Signals.DefaultSignal).
StartE <SignalName>	At the end of the currently sourcing signal's segment, the instrument sources the named signal in SMEM and repeats it continuously until a STOP microcode is executed, the instrument is reset, or another signal is started. This microcode takes an optional argument, which is the name of the signal to be sourced. If the signal name is not specified, the default signal is used (see TheHdw.MWMultiToneSource.Pins.Signals.DefaultSignal). If the MWMultiToneSource is being used to source two sample based Signals, then each AWG has an independent end ("E"). The next sample based waveform will start when its AWG completes the waveform it is currently sourcing. When the two AWGs are operating with a non 1:1 waveform length, the next sample based waveforms will start at different times.
Start1 <SignalName>	The instrument immediately starts sourcing (on the next available logical analog clock) the named signal in SMEM and sources it exactly once. This microcode takes an optional argument, which is the name of the signal to be sourced. If the signal name is not specified, the default signal is used (see TheHdw.MWMultiToneSource.Pins.Signals.DefaultSignal).
Start1E <SignalName>	At the end of the currently sourcing signal's segment, the instrument sources the named signal in SMEM and sources it exactly once. This microcode takes an optional argument, which is the name of the signal to be sourced. If the signal name is not specified, the default signal is used (see TheHdw.MWMultiToneSource.Pins.Signals.DefaultSignal). If the MWMultiToneSource is being used to source two sample based Signals, then each AWG has an independent end ("E"). The next sample based waveform will start when its AWG completes the waveform it is currently sourcing. When the two AWGs are operating with a non 1:1 waveform length, the next sample based waveforms will start at different times.

Stop	The instrument stops sourcing the currently sourced signal immediately. STOP does not impact a signal that is running once.
StopE	The instrument stops sourcing the currently sourced signal at the end of the signal. If the MWMultiToneSource is being used to source two sample based Signals, then each AWG has an independent end ("E"). Each AWG will stop when it completes the waveform it is currently sourcing. When the two AWGs are operating with a non 1:1 waveform length this means the stopping point of the sample based waveforms will occur at different times.
LoadSettingsConnect <SignalName>	Instructs an MWMultiToneSource to load the settings for the named capture signal and connects the hardware to the DUT. SignalName is required.
Disconnect	Instructs an MWMultiToneSource to reset all connections to their disconnected state.
Enable_Inst_Cond	Enables the instrument condition. The MWMultiToneSource asserts the instrument condition to the pattern generator at the completion of a capture. A MWMultiToneSource only supports this when using a sample based (for example, modulated) Signal on one or both of its paths. If a sample based Signal is being used on both paths and COND is being set on a specific sample, then if different sample rates are involved the COND is set by the AWG which reaches that sample first.
Disable_Inst_Cond	Disables an MWMultiToneSource from asserting the instrument condition.

Notes on MWMultiToneSource Microcodes

Note the following when using microcodes with an MWMultiToneSource:

- IG-XL has rules regarding the necessary spacing in vectors between microcodes on a specific channel. If you place microcodes too close together, IG-XL will return an error. General guidelines for spacing microcodes include the following:

- Resync: <1s

- LoadSettingsConnect: 400s (typical)

- Disconnect: <200s (typical)

- Start: <10s

- An MWMultiToneSource does not use PSets. It provides the equivalent capability using Signals programming in VBT.

- When using an UltraWave logical instrument (such as an MWMultiToneSource), any vector that contains a POP microcode must be fully defined across all defined logical instrument instances.

- IG-XL returns an error if you do not follow this guideline.

- Reapplying the same Signal to a given logical instrument instance results in the hardware NOPing the application. (This prevents glitches.)

For more information and an example, see [Using UltraWave24 Instruments with POP](#).

Use of Resync

The Resync microcode in the UltraWave24 subsystem performs additional tasks when compared to the Resync used with the BBAC and VHFAC instruments. The primary function of Resync for BBAC and VHFAC is to align the analog clock for the analog instrument with the digital clock supplying the timing reference for the pattern.

For the UltraWave24 instruments, the Resync microcode also:

- Forces alignment of the internal AWG clocks which drive I and Q modulation AWGs

- Synchronizes the start of waveforms

The instantiation time associated with Resync is 1 s, which you must manage. Therefore, you must place a wait of at least 1 s between the Resync and following microcode.



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UltraWave24 MWMultiToneSource Programming : MWMultiToneSource Pattern Control :
MWMultiToneSource ASCII Pattern Syntax

MWMultiToneSource ASCII Pattern Syntax

MWMultiToneSource pattern microcodes are instructions that you can place on an individual vector of a pattern. For that vector, a microcode instructs the MWMultiToneSource to perform a particular action, such as starting a new signal.

You can only use MWMultiToneSource microcodes with pins that the instruments statement has specified as MWMultiToneSource. For example:

```
instruments = {
    DUT_AIN:MWMultiToneSource;
}
```

For more information, see [Instruments Statement](#) in the Pattern Language section.

If a vector contains any MWMultiToneSource related microcode other than nop, that vector is placed in SRM.

The format for specifying an MWMultiToneSource microcode on a particular vector is:

(pin-list : MWMultiToneSource = microcode)

The following code shows an example of ASCII pattern microcode syntax for a MWMultiToneSource. The figure shows this pattern in PatternTool.

```
instruments = {
    Source:MWMultitoneSource 1;
}
```

vm_vector

```
vm_MWMultiToneSourceOnly_A ( $tset , Dig1)
{
    start_label MWMultiToneSourceOnly:
```

[illegible]

}

MWMultiToneSourceOnly.PAT (vm_MWMultiToneSourceOnly_A - VM)						
TSet	Command	Label	Vector	Source (MWMultitoneSource)	Digital	Comment
BasicSource		start_label MWMultiToneSourceOnly:	0		0	
BasicSource			1	Resync	0	
BasicSource	repeat 10		2		0	1uS wait time
BasicSource			3	LoadSettingsConnect GSM	0	
BasicSource	repeat 10000		4		0	1mS wait time
BasicSource			5	Start GSM	0	
BasicSource	repeat 100		6		0	10uS wait time
BasicSource			7	LoadSettingsConnect WCDMA	0	
BasicSource	repeat 10000		8		0	1mS wait time
BasicSource			9	Start WCDMA	0	
BasicSource	repeat 100		10		0	10uS wait time
BasicSource			11	Stop	0	
BasicSource	repeat 100		12		0	10uS wait time
BasicSource			13	Disconnect	0	
BasicSource	repeat 2000		14		0	200uS wait time
BasicSource			15		0	
BasicSource			16		0	

Vertical: 65 Vectors Vectors 0 to 16

MWMultiToneSource Pattern Microcodes Example

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UW24MultitoneSrc_TOO

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UltraWave24 MWMultiToneSource Theory of Operation

UltraWave24 MWMultiToneSource Theory of Operation

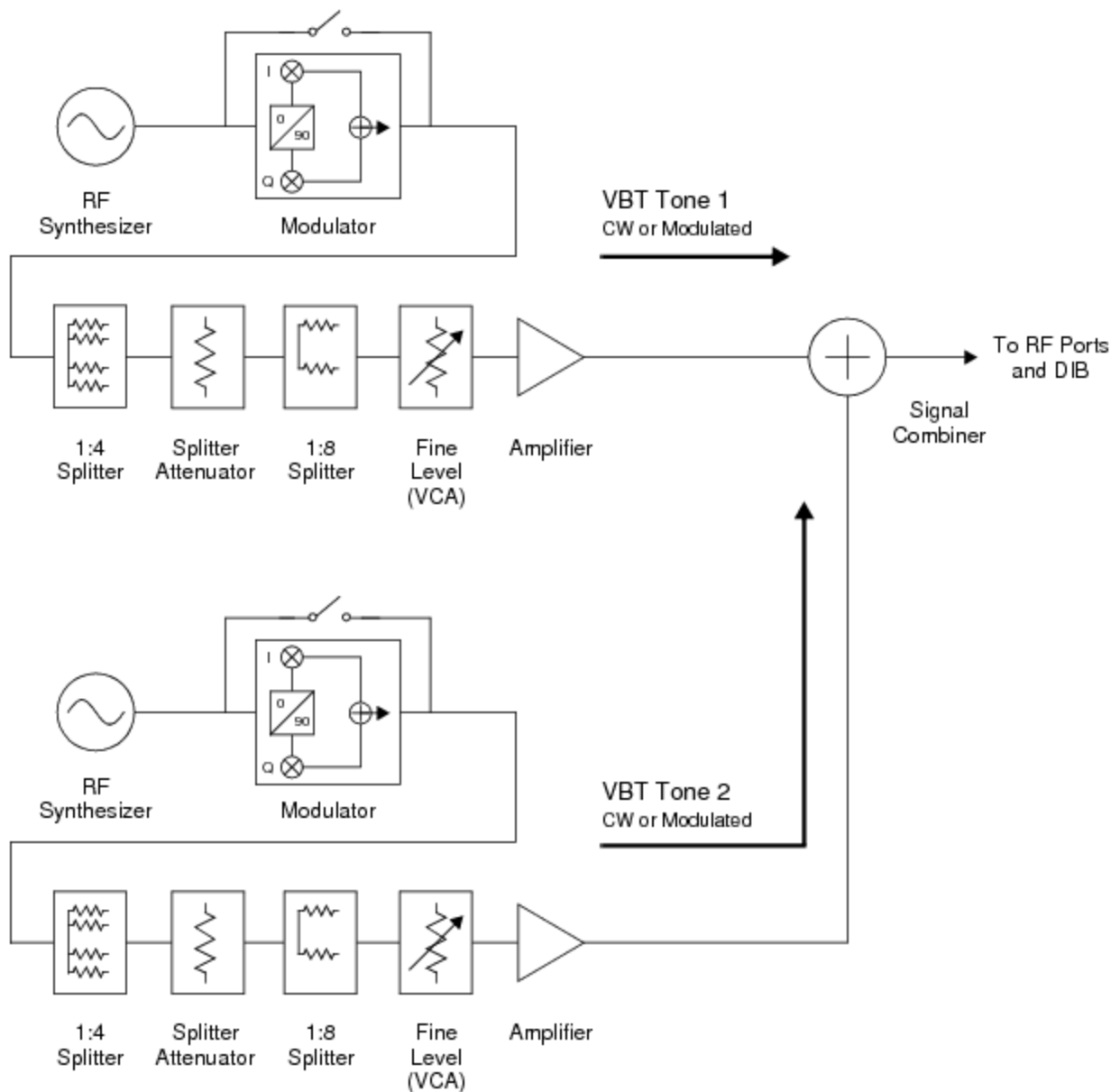
The MWMultiToneSource instrument produces a signal with two or more tones (signal frequencies) at programmable frequencies. The multitone signal is created by combining the outputs of two sources: each can be a MWSource or a MWModulatedSource. The following topics are available:

- Basic Operation
- MWMultiToneSource Hardware Functions and Signal Flows

For information on MWModulatedSource instrument performance specifications and standards, see UltraWave24 Specifications.

Basic Operation

Each of the two source instruments in a MWMultiToneSource generates signals in the same way as an individual CW or modulated source. However, the outputs of the two sources (the tone and the subtone) are routed through a signal combiner, or add path, function. See the figure.



UltraWave24 MWMultiToneSource Block Diagram

Note:

The signal path of a MWMultiToneSource has more than one separate attenuator and amplifier. All level control elements are combined in the blocks in the diagram.

The figure represents a CW source with a switched bypass path around the modulator. For both a CW and modulated signal, the signal path characteristics before the signal combiner are different for the two sources.

Specifications for the paths are included in the UltraWave Specification. For a diagram of the paths, see MWMultiToneSource Hardware Functions and Signal Flows.

For a description of the operation of an individual CW source, see CW Source Basic Operation in the MWSource instrument manual.

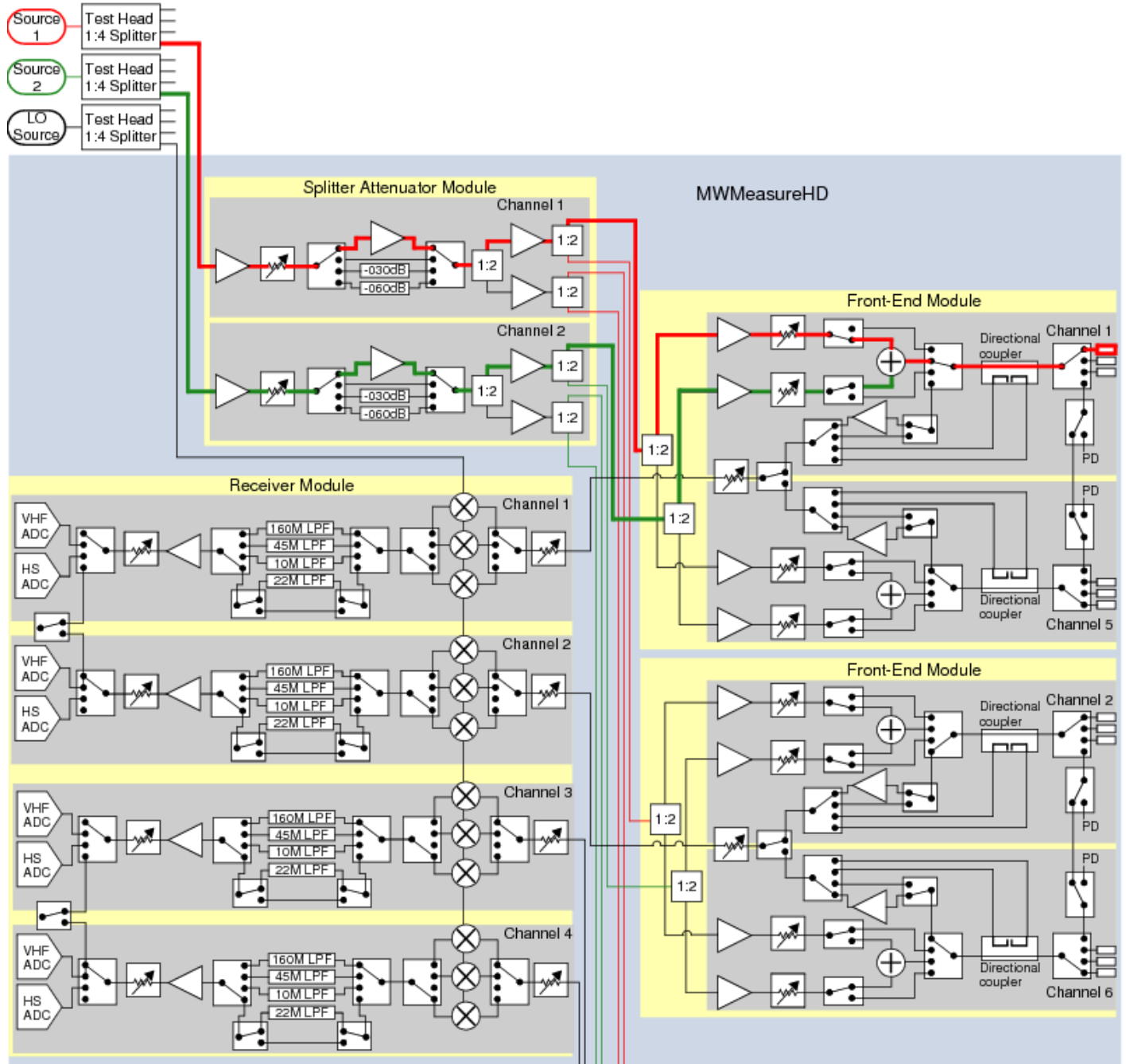
For a description of the operation of an individual modulated source, see Basic Operation in the MWModulatedSource instrument manual.

MWMultiToneSource Hardware Functions and Signal Flows

The figure shows the configuration of the UltraWave24 subsystem and source signal flow for the operation and connection of a MWMultiToneSource adding two CW signals. Heavy lines represent source signal flows. The tone and subtone sources and paths are labeled.

The two sources do not share the same RF synthesizer. The MWMultiToneSource uses two separate CW/modulated sources, with an RF synthesizer for each one. However, if there are MWMultiToneSources on more than one MWMeasureHD board, they share RF Synthesizers.

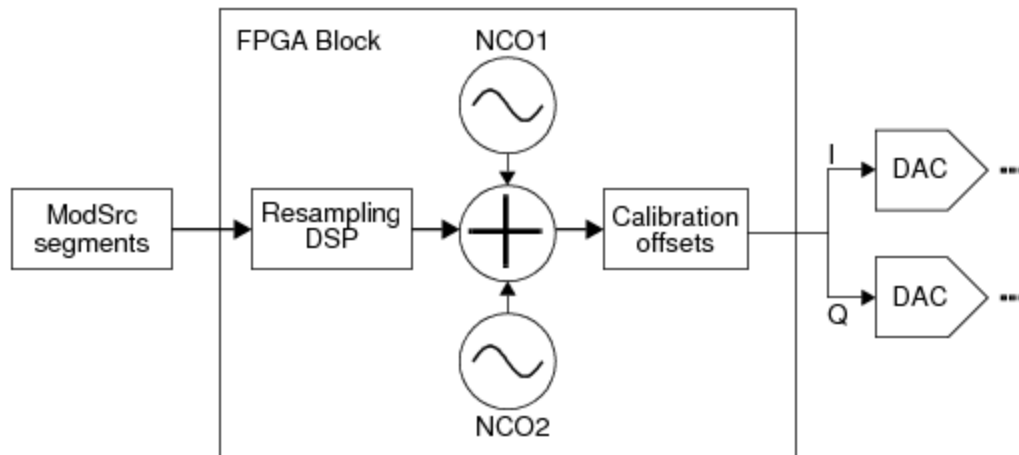
See [UltraWave24 Hardware Functions](#) for more information about the functions in the diagram.



MWMultiToneSource Hardware Functions and Signal Path, Two of Four FEMs Shown
 For DIB connections (MW ports, Pogo pins) and signal names for the MW subsystem and instruments, see [User View of UltraWave24 SMPM Connector Blocks](#), [User View of UltraWave24 DIB Pad Patterns](#), and [UltraWave24 Signal Descriptions](#).
[Creating Subtones](#)

The UltraWave24 includes the ability to generate subtones. Subtones are digitally generated CW tones that are summed with the standard CW or modulated signals from the UltraWave 24 sources. Subtones are presented under the MWMultiToneSource logical instrument.

UltraWave24 architecture allows you to use subtones as blockers in arbitrary setups or in a tritone stimulus, while using only one RF synthesizer. The following diagram shows how subtones are used.



UltraWave24 MWMultiToneSource Block Diagram

The signal flow is as follows:

- You create MWModulatedSource segments independently from the subtones. These segments are stored as complex segments in tester bulk memory.
- The ModSrc segments are sent to the resampling DSP in the UltraWave24 hardware. The DSP converts the F_s that you specify to a fixed DAC F_s .
- The resulting subtone signal is input to the summing node along with numerically controlled oscillators (NCO1 and NCO2). The summing node has independent gain control for each of the three inputs and the output.
- Calibration offsets are applied. The primary tone (if it is CW) is supplied by the ModSrc's local oscillator (LO). DC null is intentionally shifted.
- I and Q signals are produced using the DACs and sent to the device pins. The downstream RF level is adjusted such that the power at the test port remains accurate.

For more information, see [Using Subtones](#) in the Programming section.

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